Quiz 01

Name: _____ Group:_____

1. What is Cisco Packet Tracer?

A) A physical router B) A network simulation tool C) An operating system

2. Which workspace in Packet Tracer is used to build and view the network topology?

A) Physical Workspace B) Logical Workspace
C) Command Prompt D) Event List

3. Which mode shows normal operating network behaviour in Packet Tracer?

A) Realtime mode B) Simulation mode C) Debug mode D) Offline mode

4. Which mode allows step-by-step observation of packet movement?

A) Realtime mode B) Simulation mode C) Configuration mode D) Setup mode

5. Which command is used to test network connectivity between the PCs?

A) ipconfig B) ping C) traceroute D) netstat

6. What path does a packet follow when PC0 communicates with PC1?

A) PC0 → PC1 B) PC0 → Switch0 → PC1
C) PC0 → Router0 → PC1 D) PC0 → Hub0 → PC1

7. What indicates that a connection link is active in Packet Tracer?

A) Link indicator turns red B) Link indicator turns green
C) Link indicator turns yellow D) Link indicator flashes blue

8. Why are both PCs configured with the same subnet mask?

A) They are on different networks B) They belong to the same network
C) To speed up file transfer D) To disable security rules

9. What type of address is an IP address?

A) Physical address B) Logical address C) MAC address D) Port address

10. How many total bits are in an IPv4 address?

A) 16 B) 32 C) 64 D) 128

11. Into how many octets is an IPv4 address divided?

A) 2 B) 4 C) 8 D) 16

12. How many bits are contained in a single octet or byte?

A) 4 B) 8 C) 16 D) 32

13. What is the maximum possible decimal value for an IPv4 octet?

A) 128 B) 254 C) 255 D) 256

14. What binary string corresponds to the maximum octet decimal value of 255?

A) 00000000 B) 10101010 C) 11110000 D) 11111111

15. What is the binary place value of the leftmost bit in an 8-bit octet?

A) 64 B) 128 C) 256 D) 512

16. What is the decimal equivalent of the binary octet 11000000?

A) 168 B) 192 C) 224 D) 240

17. What is the binary representation of the decimal number 10?

A) 00001010 B) 00000101 C) 00010010 D) 00100100

18. What is the decimal value of the binary octet 10101000?

A) 160 B) 168 C) 192 D) 200

19. Which notation format is normally used to write IPv4 addresses?

A) Hexadecimal B) Dotted decimal C) Pure binary D) Octal

20. What are the two logical parts that make up an IPv4 address?

A) MAC and IP B) Host and Port C) Network and Host D) Subnet and Gateway

21. Which element determines the division between the network portion and host portion?

A) Default gateway B) Subnet mask C) MAC address D) Switch

22. What is the decimal representation of a /24 subnet mask?

A) 255.0.0.0 B) 255.255.0.0 C) 255.255.255.0 D) 255.255.255.255

23. How many bits are designated for the network portion in a /24 subnet mask?

A) 8 B) 16 C) 24 D) 32

24. For the IP address 192.168.10.25 on a /24 network, what is the network address?

A) 192.168.10.0 B) 192.168.10.1 C) 192.168.10.255 D) 192.168.0.0

25. What is the broadcast address for the 192.168.10.0/24 network?

A) 192.168.10.0 B) 192.168.10.1 C) 192.168.10.254 D) 192.168.10.255

26. What is the usable host IP address range for a basic /24 network like 192.168.10.0/24?

A) 192.168.10.0 – 192.168.10.255
B) 192.168.10.1 – 192.168.10.254
C) 192.168.10.1 – 192.168.10.255
D) 192.168.10.0 – 192.168.10.254

27. Why do PCs in Department A and Department B fail to ping each other through the same switch?

A) The switch is powered off
B) The network cables are disconnected
C) They belong to two different IP networks
D) The subnet masks are incorrect

28. What network device is required to enable communication between two different IP networks?

A) Switch B) Router C) Hub D) Repeater

29. Which command is used to enable/activate a router interface?

A) interface up B) no shutdown C) enable link D) start interface

30. What command is used to verify the operational status and IP addresses of router interfaces?

A) show ip route
B) show ip interface brief
C) display interface status
D) show network config

31. What is the subnet mask for a /25 network?

A) 255.255.0.0 B) 255.255.255.0 C) 255.255.255.128 D) 255.255.255.192

32. What is the subnet mask for a /26 network?

A) 255.255.255.128 B) 255.255.255.192 C) 255.255.255.224 D) 255.255.255.240

33. How many subnets are created when borrowing 2 host bits?

A) 2 B) 4 C) 8 D) 16

34. What is the formula for calculating usable hosts per subnet?

A) $2^{8-borrowedbits}$ B) $2^{hostbits} - 2$ C) $2^{borrowedbits}$ D) $2^{networkbits} - 2$

35. How many usable hosts are in a /27 subnet?

A) 14 B) 30 C) 32 D) 62

36. What is the subnet mask for a /30 network?

A) 255.255.255.240 B) 255.255.255.248
C) 255.255.255.252 D) 255.255.255.255

37. What is the primary function of a broadcast address?

A) Identify the network ID
B) Assign IP to the default gateway
C) Communicate with all hosts in the subnet
D) Route packets across external networks

38. What is the block size for a 255.255.255.192 subnet mask?

A) 32 B) 64 C) 128 D) 256

39. What is the first usable host IP for 192.168.10.0/24?

A) 192.168.10.0 B) 192.168.10.1 C) 192.168.10.2 D) 192.168.10.254

40. Which subnet does 192.168.10.75/26 belong to?

A) 192.168.10.0/26 B) 192.168.10.64/26
C) 192.168.10.128/26 D) 192.168.10.192/26

41. How many host bits remain in a /26 prefix?

A) 4 B) 5 C) 6 D) 8

42. How many usable hosts are available in a /28 subnet?

A) 6 B) 14 C) 16 D) 30

43. How many usable hosts are available in a /30 subnet?

A) 2 B) 4 C) 6 D) 8

44. What is the process of dividing a larger network into smaller networks called?

A) Routing B) Subnetting C) Switching D) Masking

45. In traditional fixed-length subnetting, what is true about all subnets?

A) They have different sizes
B) They have the same size
C) They use variable masks
D) They overlap

46. What does VLSM stand for?

A) Very Large Subnet Masking
B) Variable Length Subnet Masking
C) Virtual Local Subnet Management
D) Variable Level Subnetting Mode

47. Which prefix provides exactly 14 usable hosts?

A) /26 B) /27 C) /28 D) /29

48. Which prefix provides exactly 6 usable hosts?

A) /28 B) /29 C) /30 D) /27

49. When allocating subnets using VLSM, which network requirement should be allocated first?

A) The smallest network
B) The largest network
C) Any random network
D) The network with the lowest IP

50. Why are two addresses subtracted when calculating usable hosts in a subnet?

A) Reserved for network and broadcast addresses
B) Reserved for DNS and DHCP servers
C) Reserved for primary and secondary switches

D) Reserved for external gateway routing